



UNIT CODE	HF2K
UNIT NAME	WEB COMPONENT DEVELOPMENT II
ASSESSMENT TYPE	WEB APPLICATION DEVELOPMENT PROJECT, RECORDED SOLUTION DEMONSTRATION, AND LIVE VIVA VOCE EXAMINATION

ASSESSMENT GUIDE

OVERVIEW

This assessment evaluates the student's ability to analyse, design, develop, implement, test, explain, and defend a JSP-based Java EE Web Application by applying advanced Web Component Development concepts, including Expression Language (EL), JavaBeans, JSP Standard Actions, JSTL, Custom Tags, and Java EE design patterns, within a given business scenario.

Students are required to:

- Analyse the given business requirements.
- Design and develop a complete Java EE Web Application.
- Apply Expression Language (EL) and JSP Standard Actions appropriately.
- Develop and integrate JavaBeans within JSP pages.
- Implement JSP Standard Tag Library (JSTL) and Custom Tags.
- Apply suitable Java EE design patterns within the application architecture.
- Submit a recorded solution demonstration.
- Attend a Live Viva Voce Examination and Project Defence.

The assessment is designed to evaluate both theoretical understanding and practical Web Component Development skills while verifying the authenticity and ownership of submitted work.

OVERALL ASSESSMENT WEIGHTING

COMPONENT	MARKS
Component A – Web Application Development Project	50
Component B – Recorded Solution Demonstration	20
Component C – Live Viva Voce Examination and Project Defence	30
TOTAL	100 MARKS

IMPORTANT NOTE

Assessment marks will be awarded based on the student's ability to:

- Analyse business requirements and design an appropriate JSP-based web application.
- Correctly implement Expression Language (EL) using implicit objects and operators.
- Develop reusable JavaBeans and integrate them using JSP Standard Actions.
- Apply JSP Standard Tag Library (JSTL) effectively.
- Design and implement Custom Tags and Tag Files according to Java EE standards.

- Apply suitable Java EE design patterns to improve application structure and maintainability.
- Demonstrate understanding of JSP architecture, page processing, and component interaction.
- Explain, justify, and defend submitted work.
- Demonstrate ownership and authenticity throughout the assessment process.

SCENARIO

BookVerse Online Library Portal

BookVerse is a rapidly growing digital library service that provides members with convenient access to a wide range of books, journals, magazines, and educational resources through an online platform.

The organisation currently manages book catalogues, member registrations, book reservations, borrowing records, reading history, and library announcements using multiple independent systems. As the number of members and resources continues to increase, the existing processes have become difficult to maintain and lack consistency across the application.

To improve operational efficiency, enhance user experience, and promote code reusability, BookVerse has decided to develop a web-based Online Library Portal using JavaServer Pages (JSP) technologies.

The proposed solution should provide a user-friendly interface that enables members and library staff to manage library services efficiently while demonstrating modern Java EE Web Component Development practices.

Students are required to analyse the business requirements and develop a JSP-based Java EE web application that demonstrates the effective use of Expression Language (EL), JavaBeans, JSP Standard Actions, JSP Standard Tag Library (JSTL), Custom Tags, Tag Files, and appropriate Java EE Design Patterns.

The primary objective of the assessment is to produce a modular, reusable, maintainable, and standards-compliant web application that follows recognised Java EE development practices. The emphasis is placed on demonstrating advanced JSP concepts rather than implementing complex business logic.

Students may use JavaBeans, collections, or other suitable in-memory data structures to manage application data. A database management system may also be incorporated where appropriate; however, database implementation is not the primary focus of this assessment.

Reasonable assumptions may be made where necessary. All assumptions must be clearly documented.

COMPONENT A – WEB APPLICATION DEVELOPMENT PROJECT

OVERVIEW

Students are required to develop a complete Java EE Web Application for the given scenario.

DEVELOPMENT REQUIREMENTS

TASK	REQUIREMENTS
Requirements Analysis	Identify system users, functional requirements, non-functional requirements, assumptions, constraints, and project scope.
Application Planning	Explain the purpose, objectives, users, application modules, and expected benefits of the proposed solution.
Application Architecture	Design an appropriate application structure showing JSP pages, JavaBeans, reusable components, custom tags, and application flow.
Expression Language (EL)	Demonstrate the use of JSP Expression Language with implicit objects including pageScope, requestScope, sessionScope, param, paramValues, header, headerValues, cookie, initParam, and pageContext. Apply both property (.) and collection ([]) access operators appropriately.
JavaBeans Integration	Develop JavaBeans and integrate them using jsp:useBean, jsp:getProperty, and jsp:setProperty with suitable attributes.
JSP Standard Actions	Implement jsp:include, jsp:forward, jsp:param, and jsp:attribute where appropriate to improve modularity and maintainability.
JSP Standard Tag Library (JSTL)	Use JSTL Core Tags to simplify page development and eliminate unnecessary Java scripting elements.
Custom Tags and Tag Files	Develop reusable custom tags or tag files that encapsulate reusable presentation logic according to Java EE standards.
Design Pattern Implementation	Apply appropriate Java EE design patterns (e.g., Model-View-Controller, Front Controller, Business Delegate, Service Locator, Transfer Object, or Intercepting Filter) and justify their use.
Testing and Evaluation	Test all major application functionality and document testing outcomes.
Future Improvements	Briefly discuss possible enhancements, maintainability considerations, and scalability improvements.

MINIMUM APPLICATION REQUIREMENTS

The developed application shall include:

- Minimum 8 JSP Pages
- Minimum 2 JavaBeans
- Expression Language (EL) implemented throughout the application
- Use of all major JSP implicit objects where appropriate
- Property (.) and Collection ([]) operators
- `jsp:useBean`
- `jsp:getProperty`
- `jsp:setProperty`
- `jsp:include`
- `jsp:forward`
- `jsp:param`
- `jsp:attribute`
- JSP Standard Tag Library (JSTL Core)
- Minimum 2 Custom Tags or Tag Library Files
- At least one recognised Java EE Design Pattern (Model-View-Controller is mandatory; additional patterns are encouraged)
- Appropriate use of application, session, request, and page scopes
- Error handling and user-friendly validation
- Consistent navigation across all pages
- Well-structured and documented source code

SUBMISSION FORMAT

SECTION	CONTENT REQUIREMENTS
Cover Page	Project Title, Student Name, Student ID, Batch, Unit Name, Submission Date
1. Introduction	Overview of the selected scenario and proposed solution
2. Requirements Analysis	Users, business requirements, assumptions, constraints
3. Application Planning	Objectives, scope, application modules, expected benefits
4. System Architecture	Overall application architecture and design
5. JSP Application Structure	Folder structure, reusable components, page navigation
6. Expression Language Implementation	Explanation of EL usage, implicit objects, operators, and examples
7. JavaBeans Implementation	JavaBeans design and integration using Standard Actions

8. JSP Standard Actions	Explanation of implemented JSP Standard Actions
9. JSTL Implementation	Core JSTL tags used within the application
10. Custom Tags and Tag Files	Design, implementation, and reuse of custom tags and tag files
11. Design Pattern Implementation	Explanation and justification of selected design pattern(s)
12. Testing and Evaluation	Functional testing, screenshots, and testing evidence
13. Future Improvements	Proposed enhancements and scalability considerations
14. Conclusion	Summary of outcomes achieved
Appendices	Source Code, Screenshots, Supporting Evidence

IMPLEMENTATION EVIDENCE

CATEGORY	EVIDENCE REQUIRED
Expression Language	JSP source code demonstrating EL implementation
JavaBeans	Bean classes and integration evidence
JSP Standard Actions	JSP pages demonstrating Standard Actions
JSTL	JSP pages demonstrating JSTL implementation
Custom Tags	Tag handler classes or tag files
Design Patterns	Source code demonstrating selected pattern(s)
Testing	Functional testing evidence and screenshots
Documentation	Project report with explanations and supporting diagrams

SUBMISSION REQUIREMENTS

ITEM	FORMAT
Project Report	PDF
Source Code	ZIP
Supporting Screenshots	Included in Report
Build Instructions (if required)	Included in Report
Recommended Report Length	12–18 Pages (excluding appendices)

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Requirements Analysis and Application Planning	05
Expression Language (EL) Implementation	10
JavaBeans and JSP Standard Actions	10
JSTL Implementation	05
Custom Tags and Tag Files	10
Java EE Design Pattern Implementation	05
Testing, Documentation, and Overall Quality	05
TOTAL	50 MARKS

COMPONENT B – RECORDED SOLUTION DEMONSTRATION

OVERVIEW

Students are required to prepare and submit an individual self-recorded video presentation explaining the design, implementation, and functionality of the JSP-based web application developed for the assessment scenario.

The purpose of the recording is to demonstrate the student's understanding of the submitted solution and provide evidence of ownership prior to the Viva Voce Examination.

PRESENTATION REQUIREMENTS

Students must:

- Display their National Identity Card (NIC), Passport, Driving Licence, or other approved photographic identification at the beginning of the recording.
- Appear clearly on camera throughout the recording.
- Use their own voice throughout the presentation.
- Demonstrate personal understanding of the submitted solution.
- Clearly explain the work developed.

Medium: English

PRESENTATION CONTENT

Students should explain:

- Requirements Analysis
- Application Structure
- Expression Language (EL) Implementation
- JavaBeans Implementation
- JSP Standard Actions
- JSP Standard Tag Library (JSTL) Implementation
- Custom Tags and Tag Files
- Java EE Design Pattern Implementation
- Application Demonstration
- Key Technical Decisions

SUBMISSION FORMAT

- Video Format: MP4
- Duration: 5–10 Minutes

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Explanation of Requirements and Application Design	05
Demonstration of Expression Language, JavaBeans, and JSP Standard Actions	05
Explanation of JSTL, Custom Tags, Tag Files, and Design Pattern Implementation	05
Communication Skills, Professional Presentation, Ownership, and Authenticity	05
TOTAL	20 MARKS

COMPONENT C – LIVE VIVA VOCE EXAMINATION AND PROJECT DEFENCE

OVERVIEW

Students are required to attend a Live Viva Voce Examination and Project Defence.

The purpose of the assessment is to verify:

- Knowledge and understanding.
- Practical Web Component Development skills.
- Application of advanced JSP development concepts.
- Problem-solving ability.
- Ownership and authenticity of submitted work.
- Ability to respond to new technical requirements.

DURATION

- Approximately 15–20 Minutes

VIVA VOCE REQUIREMENTS

Following the submission of Components A and B, students may request to undertake the Viva Voce Examination through the Student Portal before the published application deadline.

Students must be prepared to:

- Demonstrate the submitted web application.
- Explain the project requirements and overall solution architecture.
- Describe the implementation of Expression Language (EL).
- Explain the use of JSP Standard Actions and JavaBeans.
- Demonstrate the implementation of JSTL Core Tags.
- Explain the design and implementation of Custom Tags and Tag Files.
- Describe the Custom Tag Event Model and PageContext API where applicable.
- Explain the selected Java EE Design Pattern(s).
- Justify technical decisions made during development.
- Defend implementation approaches.
- Respond to scenario-based technical questions.

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Knowledge and Understanding of Advanced JSP Technologies	05
Application of Web Component Development Concepts	05
Analysis of Requirements and Technical Decisions	05
Evaluation, Justification and Defence	10
Ownership and Authenticity of Submitted Work	05
TOTAL	30 MARKS

APPLICATION FOR THE VIVA VOCE EXAMINATION

Students are responsible for applying for the Viva Voce Examination through the Student Portal before the published application deadline.

Failure to apply within the published deadline will result in the student being recorded as **Absent** and the assessment attempt being deemed unsuccessful.

The student will be required to undertake a **repeat assessment**, which may be subject to **repeat assessment fees** and a maximum achievable **grade of Pass**.

Students are therefore strongly advised to apply before the published application deadline.

ACADEMIC INTEGRITY AND AUTHENTICITY

Students are expected to maintain the highest standards of academic integrity throughout the assessment process.

All submitted reports, source code, deployment configurations, demonstrations, and Viva Voce responses must represent the student's own understanding and work.

Students may be required to participate in additional verification activities, demonstrations, live coding exercises, or oral questioning to verify authenticity, ownership, and academic integrity.

Failure to demonstrate reasonable ownership and understanding of submitted work may result in academic review procedures in accordance with institutional regulations.